

CIS 371 Web Application Programming

Styles in CSS



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The Origins of Cascading Style Sheets

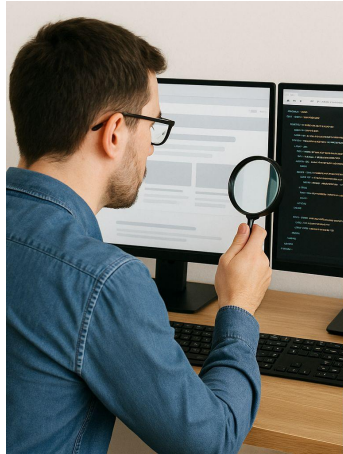
browser wars 1990s



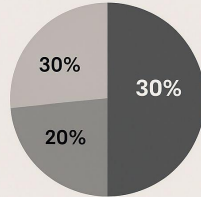
World Wide Web Consortium



universal access



CROSS-BROWSER TESTING



	Google Chrome	50%
	Mozilla Firefox	30%
	Microsoft Edge	20%

cross browser compatibility problem



HTML: structure/content CSS: presentation/style

History of CSS

20 Years of CSS

<https://www.w3.org/Style/CSS20/>

CSS Zen Garden launched in 2003. <http://www.csszengarden.com/>

Applying CSS to HTML

```
<html>                                Option 1: External
<head>
  <link rel="stylesheet"
        href="mystyles.css">
</head>
<body>
  <p>Hello world</p>
</body>
</html>
```



```
/* in mystyles.css */
p {
  border: 2px
  solid red;
}
```

```
<html>                                Option 2: Internal
<head>
  <style>
    p {
      border: 2px solid red;
    }
  </style>
</head>
<body>
  <p>Paragraph 1</p>
  <p>Paragraph 2</p>
</body>
</html>
```

```
<body>                                Option 3: Inline (Not recommended)
  <!-- inline style -->
  <p style="border: 2px solid red">.....</p>
</body>
```

Cascading Order



If we apply CSS to an HTML element using external, internal, and inline methods simultaneously, and they conflict with each other, which CSS style will take precedence?

Cascading Order

- Situation: A browser is presented with a number of CSS statements, some of which conflict with each other.
- All the styles will "cascade" into a new "virtual" style sheet by the following rules:

- | | | |
|-------------------------|---|---------|
| 1. Browser Default |  | Lowest |
| 2. External Style Sheet | | |
| 3. Internal Style Sheet | | |
| 4. Inline Style | | Highest |

- Rule 4 has the highest priority

Specific CSS rules overrule general ones.

How to define styles?

- Styles are defined using a set of rules
- Each rule
 - begins with a selector to select the element(s) onto which the rule is applied.
 - Specifies a group of properties to apply to the element(s).

```
selectorA {                                Rule 1
    property1: value;
    property2: value;
}
```

```
selectorB {                                Rule 2
    property1: value;
    property2: value;
}
```

Complete list of CSS properties:

<https://www.w3.org/Style/CSS/all-properties.en.html>

Too many CSS properties to memorize



Too many CSS properties to memorize

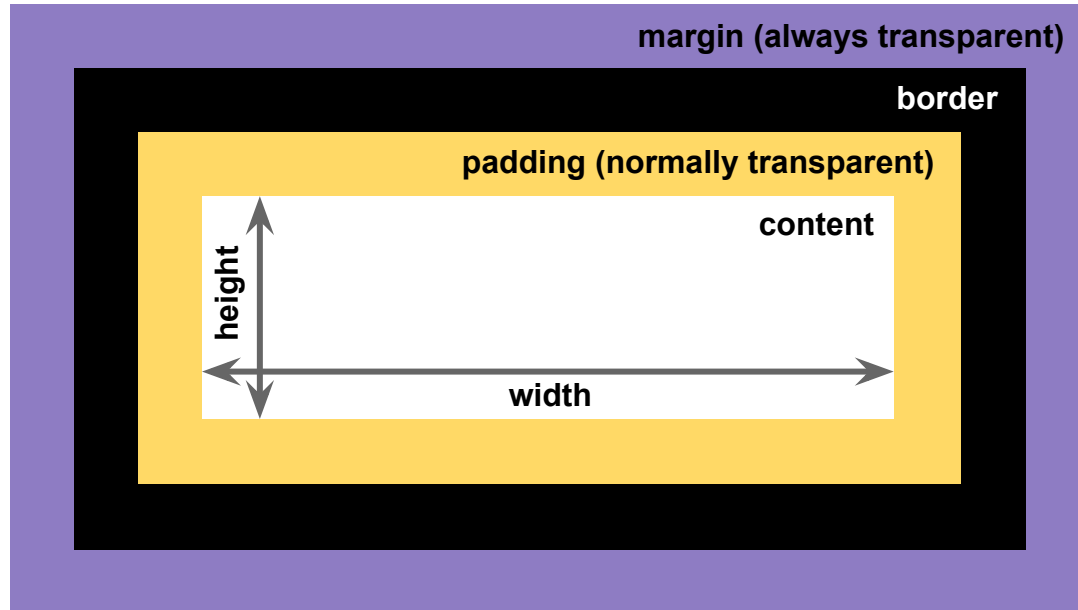


Tip

Use VSCode suggested completion to help you find what you are looking for!



CSS Box Model



background-color paints the content, padding, and border

CSS Box Model: Padding (inside the border)

```
span {  
  padding: 4px;  
  border: 12px solid green;  
  background: beige;  
}
```

```
<span>Sample Text</span>
```

Sample Text

```
span {  
  padding: 16px;  
  border: 12px solid green;  
  background: beige;  
}
```

Sample Text

CSS Box Model: Margin (outside the border)

```
span {  
  margin-right: 2px;  
  border: 8px solid green;  
  background: beige;  
}
```

```
<span>Sample</span> Text
```

Sample Text

```
span {  
  margin-right: 8px;  
  border: 4px solid green;  
  background: beige;  
}
```

Sample Text

CSS Colors

DeepSkyBlue	Gold	DarkRed
Beige	Orange	ForestGreen
HotPink	Olive	YellowGreen

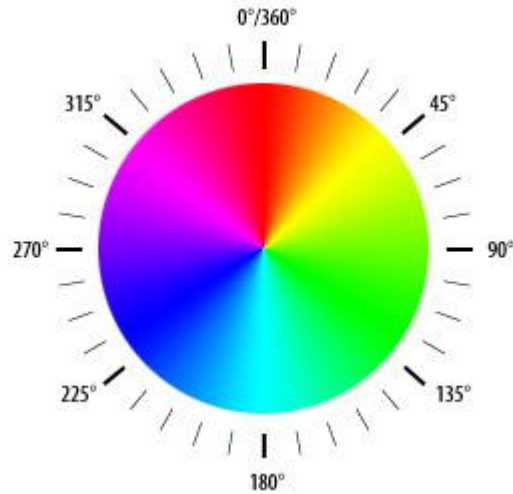
140 standard names

https://www.w3schools.com/colors/colors_names.asp

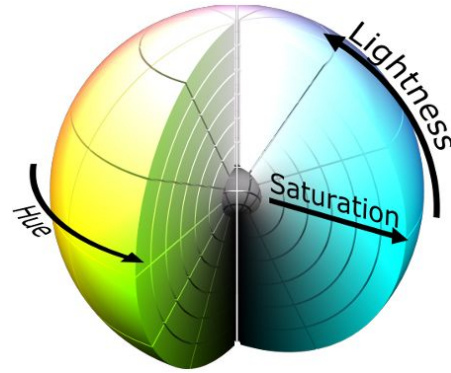
CSS Colors

- **RGB** (0-255 per color) & **Alpha Transparency**
 - `rgb(155, 138, 73)`
 - `rgba(155, 138, 73, 0.6)`
- **Hex String** (00-FF per color)
 - `#C55` or `#FCA9`
 - `#9B8A49` or `#9B8A493F`
- **HSL**
 - `hsl(20, 85%, 30%)` or `hsla(20, 85%, 35%, 0.7)`
 - Benefit: easy to generate shades of a particular color
(in code)

HSL Colorspace



Hue

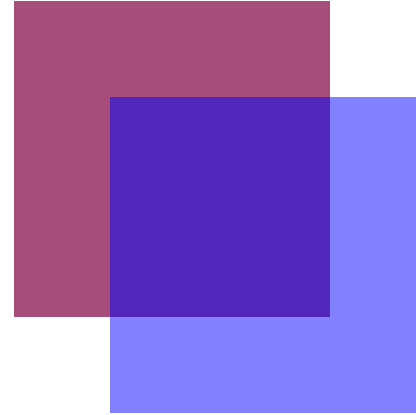
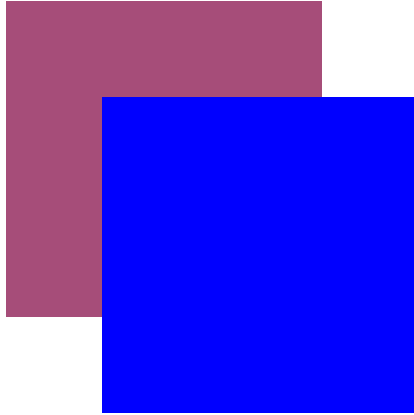


	Description	Range of values
Hue	Color Tone	Red:0, Green:120, Blue:240
Saturation	How much “ink” in your paint	0%: no ink, 100%: max ink
Lightness	How much light available when you are viewing the color	0%: no light 100%: infinite amount of light

[HSL Color Picker \(CodePen\)](#)

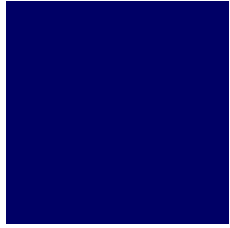
[YUI HSL Color Picker](#)

Color Transparency

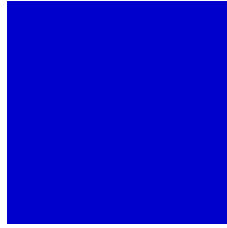


	Opaque Blue	50% Transparent Blue
RGB	<code>rgb(0, 0, 255, 1.0)</code>	<code>rgb(0, 0, 255, 0.5)</code>
Hex String	<code>#0000FFFF</code>	<code>#0000FF7F</code>
HSL	<code>hsl(240, 100%, 50%, 1.0)</code>	<code>hsl(240, 100%, 50%, 0.5)</code>

HSL Practical Use: shade of color tones



hsl(240, 100%, 20%)



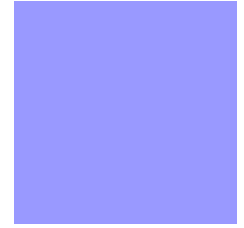
hsl(240, 100%, 40%)



hsl(240, 100%, 60%)



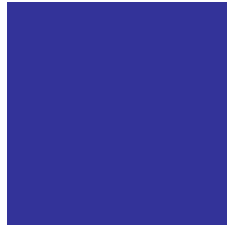
hsl(240, 100%, 70%)



hsl(240, 100%, 80%)



hsl(240, 50%, 20%)



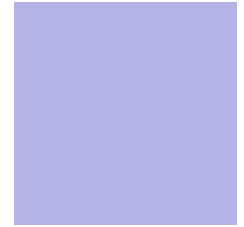
hsl(240, 50%, 40%)



hsl(240, 50%, 60%)



hsl(240, 50%, 70%)

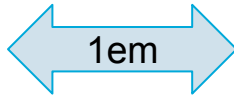


hsl(240, 50%, 80%)

Font Size: 1em

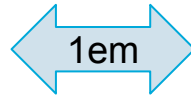
Font: Syncopate

M



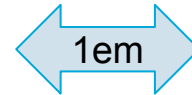
Font: Roboto

M



Font: Lobster

M



1em: the width of uppercase M in the current font (traditional interpretation)

1em: the width of the current font (modern typography interpretation)

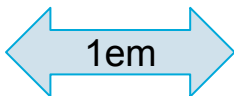
1 em: relative to the nearest parent's font

1 rem: relative to the root font

Font Size: 1em

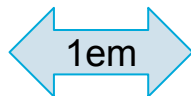
Font: Syncopate

M



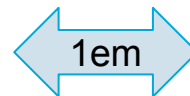
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1 em: relative to the nearest parent's font

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Use “em” for setting spacing around
your text

Applying CSS to HTML

HTML	CSS	Scope of Application
<pre><!-- by unique id --> <tag id="ticket">content</tag></pre>	<pre>#ticket { padding-left: 2em; }</pre>	Only to one element <code>#ticket</code>
<pre><!-- by tag name --> <xyz>content</xyz></pre>	<pre>xyz { font-weight: bold; }</pre>	All <code><xyz></code> tags in the document
<pre><!-- by class name --> <xyz class="weekend">content</xyz></pre>	<pre>.weekend { border: 2px solid brown; }</pre>	All <code>.weekend</code> class in the document
<pre><!-- by other attributes --> <xyz anyattr="somevalue">content</xyz></pre>	<pre>[anyattr] { background-color: white }</pre>	All tags with this attribute <code>anyattr</code> in the document

Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
#abbrev {
  border-color: red
}
```

Hello World

I am learning CSS

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span {
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</html>
```

```
[lang] {
  border-color: red
}
```

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    </p>
  </body>
</html>
```

```
[lang=de] {
  border-color: red
}
```

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  </body>
</html>
```

```
[lang=de] {
  border-color: red
}
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Hello World

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Example: Using CSS Selectors

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  </body>
</html>
```

```
#abbrev {
  border-color: red
}
```

Hello World
I am learning CSS

```
span {
  border-color: red
}
```

Hello World

I am learning CSS

```
[lang] {
  border-color: red
}
```

Hello World

I am learning CSS

```
[lang=de] {
  border-color: red
}
```

Hello World

I am learning CSS

CSS Selector Specificity

```
/* In mystyles.css */  
div {  
    background: red;  
}  
  
#top {  
    background: green;  
}  
  
.warn {  
    background: yellow;  
}
```

```
<!-- In HTML -->  
<div id="top" class="warn">  
    Sample  
</div>
```



Which selector wins?

Specificity Calculator
(Higher score wins)

CSS Selector Specificity Exercise

```
<div id="header">
  <ul class="menu">
    <li class="item">Home</li>
    <li class="item">About</li>
    <li class="item special">Services</li>
    <li class="item">Contact</li>
  </ul>
</div>
```

HTML



What is the color of the "Services"?

[Specificity Calculator](#)

CSS

```
/* Selector 1 */
ul.menu li {
  color: blue;
}

/* Selector 2 */
li.item {
  color: green;
}

/* Selector 3 */
#header .item.special {
  color: red;
}

/* Selector 4 */
#header ul li.special {
  color: orange;
}
```

Practice